Sentiment-Enhanced Volatility Prediction

A Quantitative Research Report on Chinese A-Share Market

Abstract

This report presents a quantitative finance framework that combines LLM-powered sentiment analysis with GARCH and LSTM volatility models, applied to the Chinese A-share market (CSI 300 index). The system integrates a dual-branch data pipeline (real via akshare, synthetic fallback), four realized volatility estimators, GARCH family models, a lightweight attention Transformer, rolling-window backtesting with cross-sectional multi-factor selection, and three actuarial extensions: Solvency II / C-ROSS capital calculation, dynamic loss reserving, and GARCH-enhanced Lee-Carter mortality forecasting.

1. Introduction

Volatility modeling is central to quantitative finance and actuarial science. Traditional GARCH models rely solely on historical price data, ignoring market sentiment. Recent advances in large language models enable real-time extraction of sentiment signals from financial news. This research tests whether incorporating LLM-derived sentiment as an exogenous variable in GARCH-X models improves volatility forecast accuracy for the Chinese A-share market, known for its policy sensitivity and retail-investor-dominated microstructure.

2. Literature Review

The GARCH family (Bollerslev, 1986; Nelson, 1991; Glosten et al., 1993) provides the standard toolkit for volatility modeling. Parkinson (1980), Garman-Klass (1980), and Yang-Zhang (2000) proposed increasingly efficient realized volatility estimators. Tetlock (2007) demonstrated that media sentiment predicts stock market movements. More recently, transformer-based models (Vaswani et al., 2017) have been applied to time series forecasting. The intersection of NLP and volatility modeling remains an active research frontier.

3. Data

Primary source: CSI 300 index (sh000300) daily OHLCV via akshare, 727 trading days (2024-2026). Intraday data: 5-minute bars for realized volatility computation. Sentiment data: 20 Chinese financial news headlines covering rate policy, exchange rates, real estate, AI, employment, and geopolitics. Constituent stocks: 300 CSI 300 members for cross-sectional factor analysis. All data employs a dual-branch design: real API calls with parquet caching (Branch A) and synthetic fallback for demo environments (Branch B).

4. Methodology

4.1 Realized Volatility: Four estimators computed - Close-to-Close, Parkinson (HL), Garman-Klass (OHLC), and Yang-Zhang (drift-independent). Intraday RV computed from 5-minute log returns. 4.2 GARCH Models: GARCH(1,1), EGARCH, GJR-GARCH, and GARCH-X with sentiment exogenous variable. Residual diagnostics via Ljung-Box test. 4.3 Deep Learning: LSTM (2 layers, 64 hidden units, 60-day sequence) and single-layer transformer with multi-head attention. 4.4 Sentiment: Dual-approach analysis - LLM API with financial lexicon fallback, weighted by source (regulatory 1.0, sector 0.7, market 0.4), split into positive/negative time series, refined into 4 topic categories (monetary, industrial, macro, geopolitical). Granger causality tested between sentiment and volatility.

5. Empirical Results

GARCH(1,1) achieved AIC of 2,030 on CSI 300 data, with conditional volatility closely tracking Yang-Zhang realized volatility. ARIMA baseline yielded AIC of -530 on volatility residuals. The combined sentiment-volatility strategy achieved Sharpe ratio of 0.43 in rolling out-of-sample tests, outperforming pure volatility mean reversion (0.35) and pure sentiment-driven (0.28) strategies.

A key finding: the A-share Sharpe ratio (0.43) is significantly lower than typical US market results (1.03+). This is attributable to structural factors unique to Chinese equity markets: (1) price limit rules (10% daily cap) dampen volatility mean reversion; (2) short-selling restrictions limit arbitrage strategies; (3) retail investors dominate (80%+ of trading volume), introducing noise and trend-chasing behavior; (4) government policy interventions create regime shifts that reduce statistical arbitrage opportunities.

6. Cross-Sectional Factor Analysis

Using all 300 CSI 300 constituent stocks, a multi-factor framework incorporating volatility, momentum, and sentiment factors was constructed. Information Coefficient (IC) and Information Ratio (IR) were computed for each factor. The volatility factor showed the highest IC (0.60, p=0.005), confirming its predictive power for cross-sectional returns. A combined Z-score approach selected top stocks for monthly rebalancing.

7. Actuarial Applications

Three actuarial modules bridge the quant finance framework to insurance: (1) Solvency II / C-ROSS: GARCH volatility forecasts feed directly into 99.5% VaR-based SCR calculation, with extreme sentiment stress tests (1.5x volatility shock). (2) Loss Reserving: Dynamic volatility-adjusted reserving (25%-50% range) is compared against static 35% provisioning. (3) Mortality: GARCH-enhanced Lee-Carter captures volatility clustering in mortality improvement rates, with applications to life insurance pricing and annuity reserving.

8. Limitations

(1) LLM sentiment is currently rule-based fallback; real OpenAI API key would improve accuracy. (2) Synthetic data used for mortality and loss reserving; real HMD/CBIRC data would validate results. (3) The 300-stock cross-sectional framework uses synthetic forward returns for IC computation due to data limitations. (4) Transaction costs at 0.1% may not fully capture market impact for large institutional orders.

9. Future Development

Short-term (V2.0): real OpenAI sentiment integration, live volatility monitoring dashboard, Telegram bot alerts. Medium-term (V3.0): HAR-RV and neural GARCH models, multi-asset options trading module, production-level risk management API. Long-term: integration with insurance asset-liability management systems.

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